PACKET 1 • SOFTWARE DEVELOPMENT

2 Context Changes Everything

RefView Window/Level Display Transform

Additional RefView specification

For pixel value x, window center c, and window width w:

```
lower = c - 0.5 - (w - 1) / 2
upper = c - 0.5 + (w - 1) / 2
```

```
if x <= lower:
    output = 0
else if x > upper:
    output = 255
else:
    output = ((x - (c - 0.5)) / (w - 1) + 0.5) * 255
```

Convert the calculated value to the nearest integer and constrain it to 0–255. For this workshop, w must be greater than 1; invalid widths must produce an explicit error rather than silently substituting another value.

Worked example

```
center = 50     width = 101
lower = -0.5    upper = 99.5
```

Pixel value	Expected display value
-100	0
-1	0
0	approximately 1
50	approximately 129
99	approximately 254
100	255
500	255

Your task Revisit your solution using this additional context. The acceptance criteria have not changed.

PACKET 2 • RESEARCH SCIENTISTS

2 Context Changes Everything

RefView SmoothCT Experimental Evaluation

Available experimental resources

Resource	Available information / use
Uniform phantom	Large homogeneous water-equivalent region; suitable for mean CT number and noise. Typical baseline: mean ≈ 0 HU; noise ≈ 12 HU standard deviation.
Resolution phantom	High-contrast edge suitable for estimating spatial resolution. The laboratory reports MTF50 as its primary metric.
Material phantom	Inserts with nominal CT numbers: water-equivalent 0 HU; low-density -100 HU; high-density +100 HU.
Paired images	Every SmoothCT image has the corresponding unprocessed image from the same acquisition.

Constraint

Acquisition parameters and reconstruction settings must remain identical between processed and control images.

Your task

- Revisit the experimental design using the available resources.
- Specify how each metric will be measured and compared.
- Use the paired-image structure appropriately.
- Retain the original hypothesis, thresholds, and acceptance criteria.

Reminder The hypothesis and acceptance criteria have not changed.

PACKET 3 • PRODUCT DEVELOPMENT

2 Context Changes Everything

RefView Auto-Window Capability

Additional capability description

The proposed algorithm examines the image-value histogram. For ordinary images:

```
lower display bound = 5th percentile
upper display bound = 95th percentile
window center and width are derived from those bounds
```

Product constraints

- Complete within 200 ms for a 512 × 512 image on the reference workstation.
- Produce the same result every time for identical input.
- Never prevent the user from manually changing the window.
- Fall back to a predefined default if a valid window cannot be calculated.

Known challenging images

- Large regions of air
- Extreme high-value objects
- Metal-like artifacts
- Nearly uniform pixel values
- Corrupted or missing pixel data

Your task

- Revisit the requirements, risks, and verification plan.
- Convert vague behavior into observable, measurable outcomes.
- Ensure normal, abnormal, performance, determinism, fallback, and override behavior can be verified.

Reminder The acceptance criteria have not changed.